

MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE

RESEARCH PAPER

Submitted By

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and encouragement throughout the development of this project. This internship provided me with an opportunity to apply my knowledge of **Python, data analysis, machine learning, data preprocessing, clustering techniques, and data visualization** to a practical real-world project. Working on this project helped me develop a better understanding of buyer segmentation and how machine learning can be used to generate meaningful insights from real estate data.

I am thankful to the **Unified Mentor team and my internship mentors** for their guidance, support, and valuable feedback throughout the project. Their assistance helped me understand the project requirements and improve my technical and analytical skills.

Simarpreet Kaur

Abstract

The real estate market consists of buyers with different demographic characteristics, purchasing behaviours, geographical locations, and investment preferences. Treating all buyers as a single group can reduce the effectiveness of marketing and customer targeting strategies. This project presents a machine learning-based approach for buyer segmentation and investment profiling in the real estate market.

The study uses client and property transaction datasets containing demographic information, acquisition purpose, loan behaviour, satisfaction scores, and property transaction details. Data preprocessing was performed through data cleaning, feature engineering, categorical encoding, and numerical feature scaling. Buyer-level transaction features such as number of properties purchased, total spending, average property price, and total area purchased were generated from the property transaction data.

K-Means clustering was applied to identify groups of buyers with similar characteristics. The Elbow Method and Silhouette Score were used to evaluate the appropriate number of clusters. The final K-Means model identified two major buyer segments: **Standard Property Buyers** and **High-Value Property Buyers**. The high-value segment demonstrated higher average property purchases, total spending, average property price, and total purchased area compared with the standard segment.

A Stream lit-based interactive dashboard was also developed to provide buyer segmentation, behavioural analysis, geographic analysis, and segment insights. The proposed system can help real estate organizations understand buyer behaviour and develop more targeted marketing and customer engagement strategies.

Introduction

The real estate industry is a complex and highly competitive market in which buyers have different demographic characteristics, geographical preferences, purchasing capacities, and reasons for acquiring properties. Some buyers purchase properties for personal use, while others purchase multiple properties for investment purposes. Corporate and individual buyers may also demonstrate different purchasing patterns.

Traditional approaches that treat all buyers as a single customer group may not effectively identify these differences. This can result in less targeted marketing, inefficient customer engagement, and difficulty in identifying valuable buyer groups. Therefore, understanding buyer behaviour and dividing customers into meaningful segments can help real estate organizations make better business decisions.

Machine learning provides effective techniques for discovering hidden patterns within large datasets. In particular, clustering algorithms can group customers who have similar characteristics without requiring predefined categories. This project applies machine learning techniques to real estate client and property transaction data to identify meaningful buyer segments.

The project combines demographic information, acquisition purpose, loan behaviour, satisfaction scores, and property transaction characteristics. Buyer-level features such as the number of properties purchased, total spending, average property price, and total area purchased are used to understand transaction behaviour.

K-Means clustering is used as the primary segmentation technique, while hierarchical clustering is used for comparison. The resulting buyer segments are analysed to understand their demographic, geographic, and transaction characteristics. An interactive Streamlit dashboard is also developed to present the results and allow users to explore different buyer groups using filters.

The overall aim of this project is to provide a data-driven approach to buyer segmentation that can support real estate market intelligence, customer targeting, and investment profiling.

Problem Statement

The real estate market contains buyers with different purchasing behaviours, demographic characteristics, geographical locations, and property requirements. Without proper customer segmentation, real estate organizations may use the same marketing and engagement strategies for different types of buyers.

This can lead to inefficient marketing campaigns, difficulty in identifying valuable customers, limited understanding of buyer behaviour, and missed opportunities for targeted customer engagement. There is therefore a need for a data-driven system that can identify groups of buyers with similar characteristics and transaction behaviours.

The problem addressed in this project is to develop a machine learning-based buyer segmentation system that can analyse client and property transaction data, identify meaningful buyer groups, and provide insights that can support real estate market intelligence and investment profiling.

Objectives

The main objectives of this project are:

1. To analyse demographic, geographic, and transaction-related characteristics of real estate buyers.

2. To clean and preprocess client and property transaction data for machine learning analysis.
3. To create buyer-level transaction features such as properties purchased, total spending, average property price, and total area purchased.
4. To apply K-Means clustering to identify groups of buyers with similar characteristics and purchasing behaviour.
5. To use the Elbow Method and Silhouette Score to evaluate the appropriate number of clusters.
6. To compare K-Means clustering with hierarchical clustering.
7. To interpret the characteristics and transaction behaviour of the identified buyer segments
8. To develop an interactive Stream lit dashboard for exploring buyer segments and geographic and behavioural patterns.
9. To provide data-driven insights that can support targeted marketing and real estate market intelligence.

Dataset Description

This project uses two datasets: a **Client Dataset** and a **Property Transaction Dataset**. The datasets are connected through a common client identifier, which allows client characteristics to be combined with their property transaction behaviour.

Client

Dataset:

The client dataset contains 2,000 buyer records and includes information such as client ID, client type, date of birth, gender, country, region, acquisition purpose, satisfaction score, loan application status, and referral channel.

Property

Dataset:

The property dataset contains 10,000 property records and includes information such as listing ID, tower number, transaction date, unit category, unit number, floor area, sale price, listing status, and client reference.

The property dataset was used to identify completed property transactions. Buyer-level transaction features were then created by grouping sold properties according to the corresponding client reference.

The main derived features used for buyer profiling were:

- **Properties Bought:** Number of sold properties purchased by each buyer.
- **Total Spend:** Total amount spent on sold properties by each buyer.
- **Average Property Price:** Average sale price of properties purchased by each buyer.
- **Total Area:** Total floor area purchased by each buyer.

These derived transaction features were combined with the demographic and behavioral characteristics from the client dataset to create the final buyer-level dataset used for machine learning.

Data Preprocessing

Data preprocessing was performed to prepare the datasets for analysis and machine learning. The client and property datasets were first loaded and their structures were examined to understand the available variables and data types.

The date fields were converted into appropriate datetime formats. The date of birth field was used to calculate the age of each buyer using a reference date of 31 December 2025.

The sale price field in the property dataset contained currency symbols and comma separators. These characters were removed and the values were converted into numerical format.

Duplicate records were checked in both datasets. No complete duplicate records were found.

Only properties with a **Sold** listing status were used to calculate completed buyer transaction behaviour. Available or unassigned properties were not treated as completed purchases.

The property data was grouped by client reference to calculate the number of properties purchased, total spending, average property price, and total area purchased for each buyer. These features were then merged with the client dataset using the client identifier.

Missing transaction values for buyers without calculated transaction records were filled with zero, representing no recorded completed purchase in the derived transaction features.

For machine learning, numerical variables were standardized using **Standard Scaler**, while categorical variables were converted into numerical form using **One-Hot Encoding**. This produced the final feature matrix used for clustering.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution and characteristics of the buyers and their property transaction behaviour before applying machine learning algorithms.

The analysis examined buyer age, satisfaction score, number of properties purchased, total spending, average property price, and total purchased area. Categorical variables such as client type, acquisition purpose, country, region, gender, loan status, and referral channel were also analysed.

The buyer-level dataset contains **2,000 buyers**. The average buyer age is approximately **55.60 years**, with ages ranging from approximately **25 to 94.9 years**.

A total of **7,305 completed property transactions** were identified from the property dataset. These transactions were used to calculate buyer-level purchasing behaviour.

The exploratory analysis also showed that the **United States is the most represented country** among the buyers, while California is the most frequently represented region in both identified buyer segments.

Visualizations such as bar charts and boxplots were used to compare buyer segments based on transaction characteristics. These visualizations helped identify differences in spending, number of properties purchased, average property price, and total area purchased between the buyer groups.

The EDA provided an understanding of the dataset and helped establish the features required for the subsequent machine learning-based segmentation.

Machine Learning Methodology

Machine learning clustering techniques were used to identify groups of buyers with similar characteristics and transaction behaviour. Since the project does not have predefined buyer segment labels, an unsupervised learning approach was selected.

The machine learning workflow consisted of the following stages:

1. Selection of relevant demographic, behavioural, and transaction features.
2. Encoding of categorical variables using One-Hot Encoding.
3. Standardization of numerical variables using Standard Scaler.
4. Determination of a suitable number of clusters using the Elbow Method and Silhouette Score.
5. Application of K-Means clustering as the primary segmentation algorithm.
6. Application of hierarchical clustering for comparison.
7. Analysis and interpretation of the resulting buyer segments.

The numerical features used in the clustering process included age, satisfaction score, properties bought, total spending, average property price, and total area. Categorical features included client type, gender, country, region, acquisition purpose, loan application status, and referral channel.

The resulting clusters were evaluated based on their size, demographic characteristics, purchasing behaviour, and transaction values to determine meaningful differences between buyer groups.

K-means Clustering

K-Means clustering was selected as the primary machine learning algorithm for buyer segmentation. K-Means is an unsupervised learning algorithm that groups observations into a predefined number of clusters based on similarity.

In this project, the algorithm was applied to the pre processed buyer feature matrix. The numerical variables were standardized and categorical variables were one-hot encoded before clustering so that variables with different scales could be used together.

The appropriate number of clusters was evaluated using two methods: the **Elbow Method** and the **Silhouette Score**. The Elbow Method was used to observe how the within-cluster variation changed as the number of clusters increased. The Silhouette Score was used to measure how well-separated the resulting clusters were

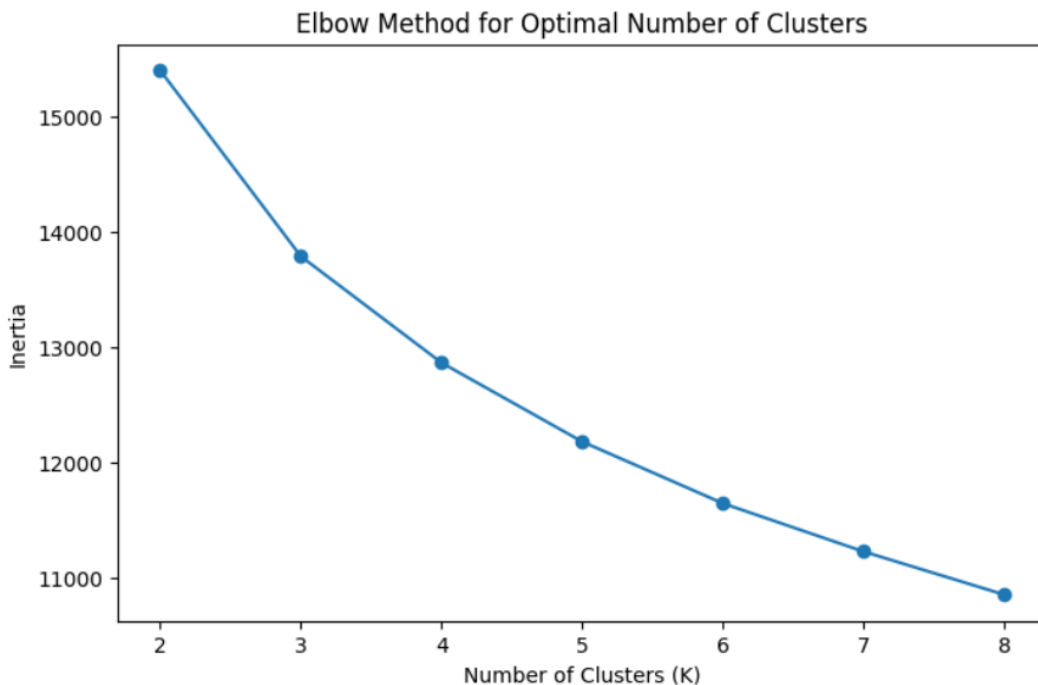


Figure 1: Elbow Method for Determining the Number of Clusters

The Silhouette Score was calculated for values of K from 2 to 8. The highest score was obtained for **K = 2, with a Silhouette Score of 0.1562**. Therefore, two clusters were selected for the final K-Means segmentation model.

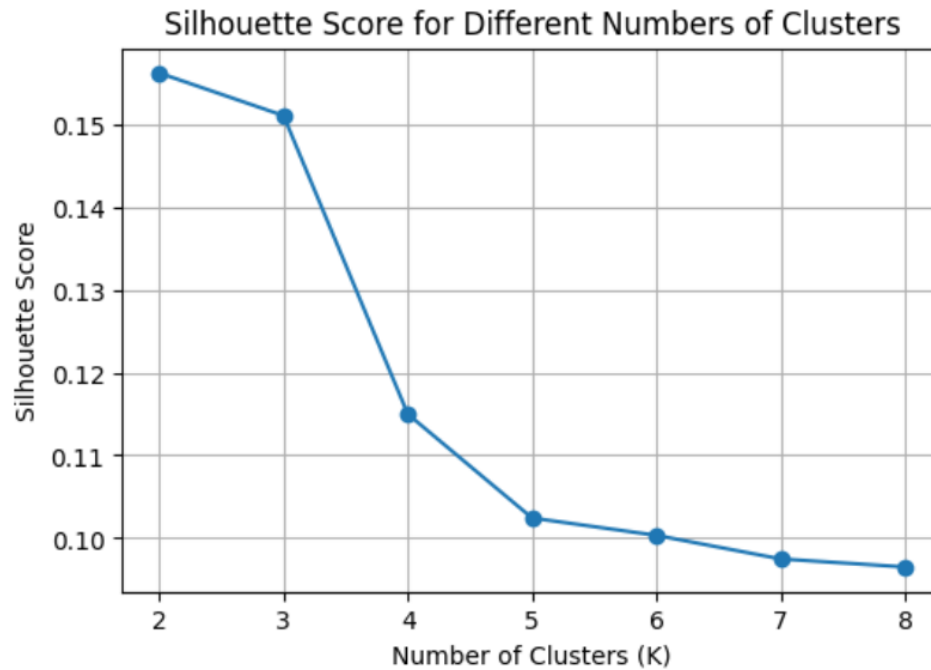


Figure 2: Silhouette Score for Different Numbers of Clusters

The final K-Means model was trained with two clusters using a fixed random state of 42 and multiple initializations to improve the stability of the clustering result.

Cluster Results And Interpretation

The final K-Means model divided the 2,000 buyers into two distinct transaction-based segments. The segments were named **Standard Property Buyers** and **High-Value Property Buyers** based on their purchasing behaviour.



Figure 3: Distribution of Buyers Across the Identified Segments

Standard Property Buyers consisted of **1,155 buyers (57.75%)**. This segment had an average of **3.39 properties purchased per buyer**, an average total spend of approximately **\$1,039,153.71**, an average property price of approximately **\$309,012.52**, and an average total purchased area of approximately **3,463.13 square feet**.

High-Value Property Buyers consisted of **845 buyers (42.25%)**. This segment had an average of **4.00 properties purchased per buyer**, an average total spend of approximately **\$1,562,755.53**, an average property price of approximately **\$399,136.62**, and an average total purchased area of approximately **5,131.56 square feet**.

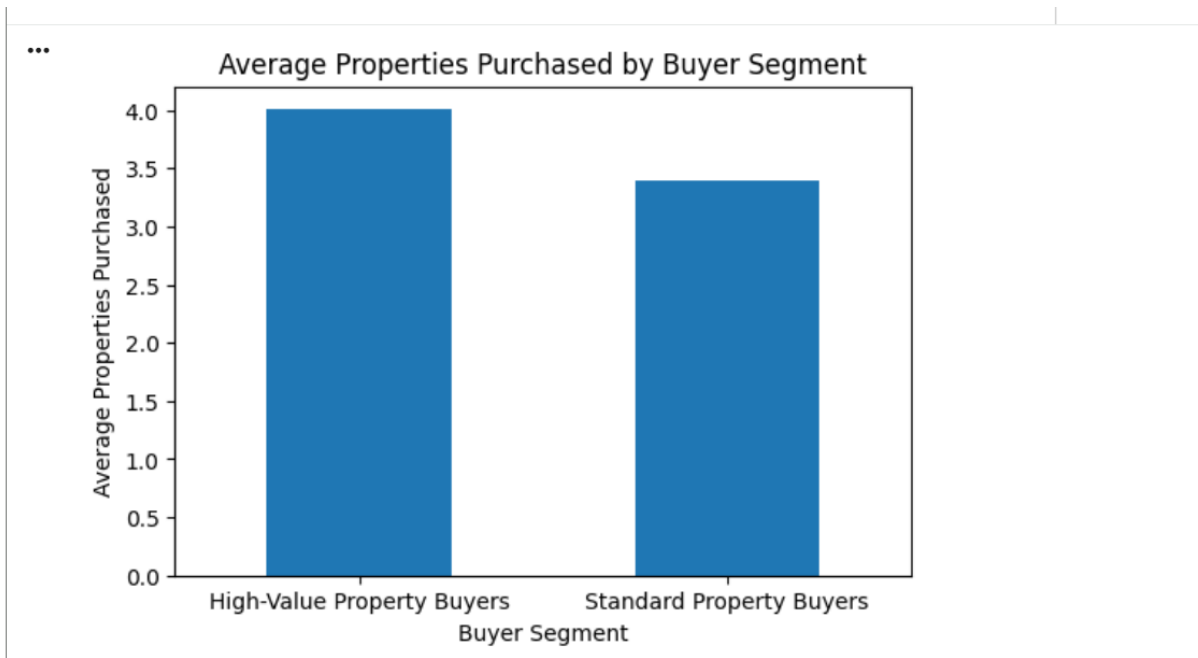


Figure 5: Average Number of Properties Purchased by Buyer Segment

The high-value segment therefore demonstrates stronger transaction activity than the standard segment. Buyers in this group purchase more properties on average, spend more in total, purchase properties with higher average prices, and acquire a larger total area.

The average age of the Standard Property Buyers was approximately **54.76 years**, while the

High-Value Property Buyers had an average age of approximately **56.75 years**. Average satisfaction scores were also similar, at approximately **2.99** for Standard Property Buyers and **3.09** for High-Value Property Buyers.

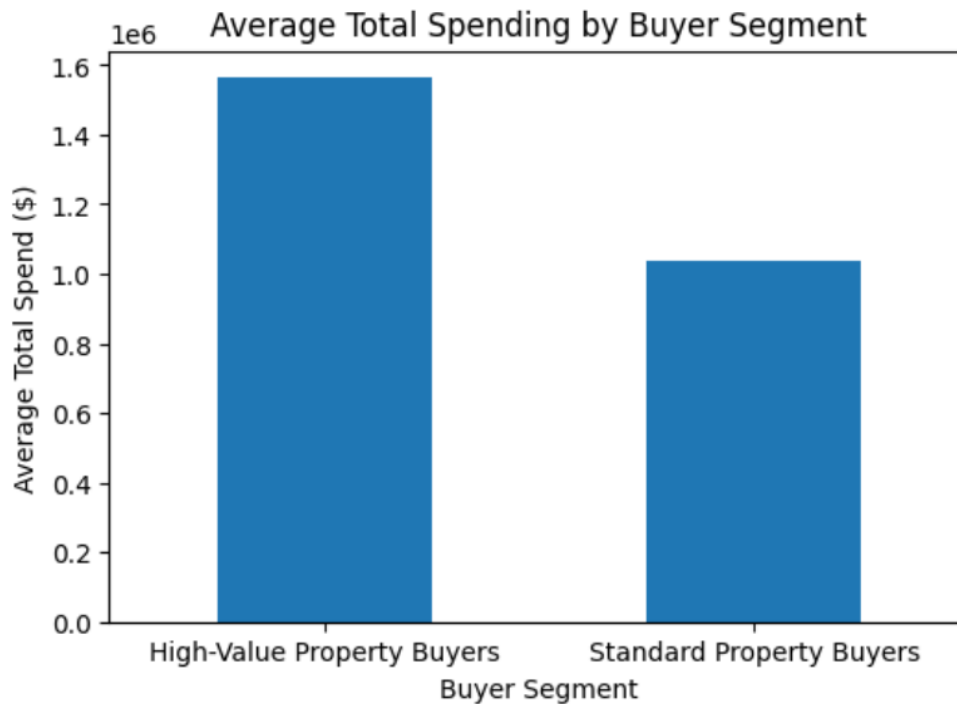


Figure 4: Average Total Spending by Buyer Segment

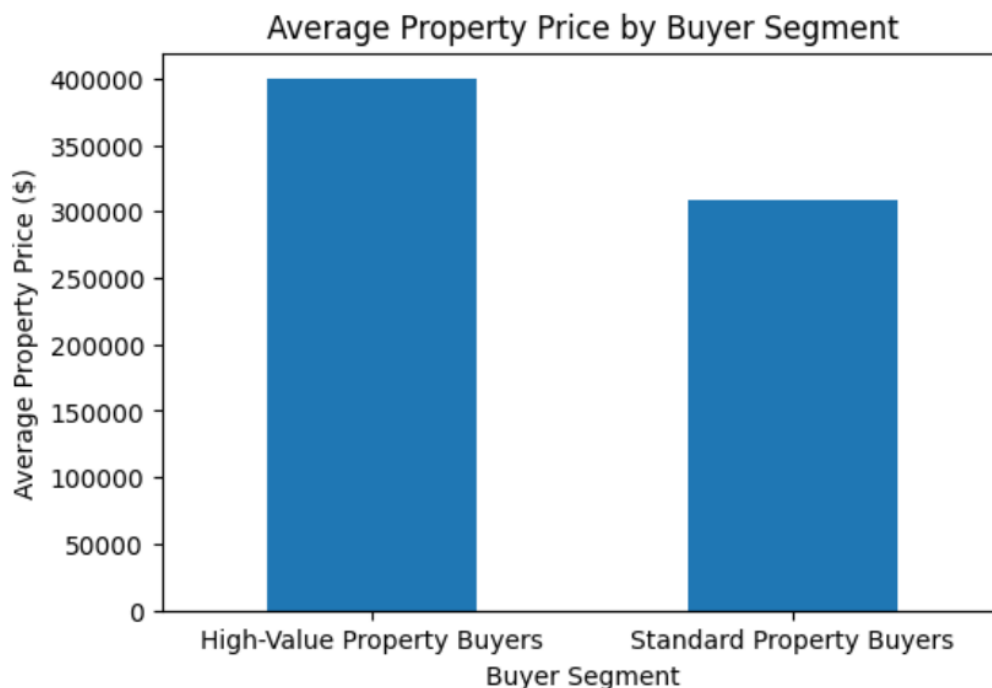


Figure 6: Average Property Price by Buyer Segment

The analysis indicates that the main distinction between the two segments is **transaction behaviour rather than demographic characteristics**. Therefore, the High-Value Property Buyers segment should be interpreted as a group with higher observed property transaction activity, rather than as a group defined by income.

Hierarchical clustering

Hierarchical clustering was evaluated as a comparison model, but K-Means achieved a higher Silhouette Score and was therefore selected as the final model.

An interactive Stream lit dashboard was developed to present the segmentation results, buyer behaviour, geographic analysis, and segment insights. The dashboard also allows users to filter the results by country, region, acquisition purpose, and client type.

Overall, the project demonstrates how machine learning and interactive analytics can be used to identify meaningful patterns in real estate buyer behaviour and support data-driven market intelligence and customer targeting. However, the identified segments should be interpreted as broad transaction-based groups because the available data does not directly measure income or complete investment capacity.

Business Insights

The buyer segmentation results provide several useful insights for real estate market intelligence.

1. Identification of high-value transaction behavior:

The High-Value Property Buyers segment represents 42.25% of the buyers and shows higher average spending, property purchases, average property prices, and total purchased area. This group can be considered an important target for personalized property recommendations and relationship management.

2. Standard buyer segment:

Standard Property Buyers represent 57.75% of the buyers. Their transaction values are lower than those of the High-Value segment, but the segment contains the majority of buyers and therefore represents an important customer base for regular marketing and engagement campaigns.

3.Acquisition purpose:

Both segments contain a larger proportion of buyers purchasing properties for home or personal use than for investment. Therefore, the identified High-Value segment should not automatically be classified as an investment-only group.

4.Geographic concentration:

The United States is the dominant country in the dataset, while California is the most represented region in both segments. This indicates that geographic targeting can be useful when designing regional marketing strategies.

5.Client type:

Individual buyers form the majority of both segments. Company buyers represent a much smaller proportion of the dataset, indicating that the identified clusters are primarily driven by individual buyer behaviour.

6.Loan behaviour:

Loan application behaviour is relatively similar between the two segments. Therefore, loan status does not appear to be a major factor distinguishing the two buyer groups.

Overall, the segmentation indicates that **transaction behaviour is the strongest practical distinction between the identified buyer groups**, while demographic and behavioural

variables such as age, gender, loan status, and acquisition purpose show less pronounced differences.

Stream lit Dashboard

An interactive web-based dashboard was developed using Stream lit to make the buyer segmentation results easier to explore and interpret. The dashboard connects the processed buyer-level dataset with interactive visualizations and filtering options.

The dashboard contains four main sections:

1. Buyer Segmentation Overview:

This section displays the total number of buyers, number of identified segments, total properties purchased, and the distribution of buyers across the two segments.

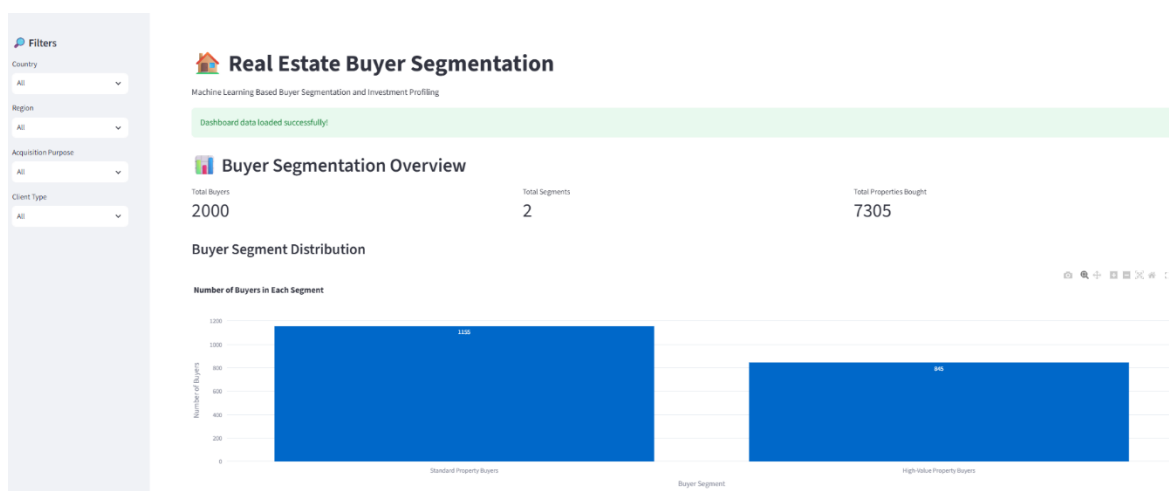


Figure 7: Stream lit Dashboard — Buyer Segmentation Overview

2. Investor/Buyer Behaviour Dashboard:

This section compares the segments based on average total spending, average number of properties purchased, and average total area purchased.

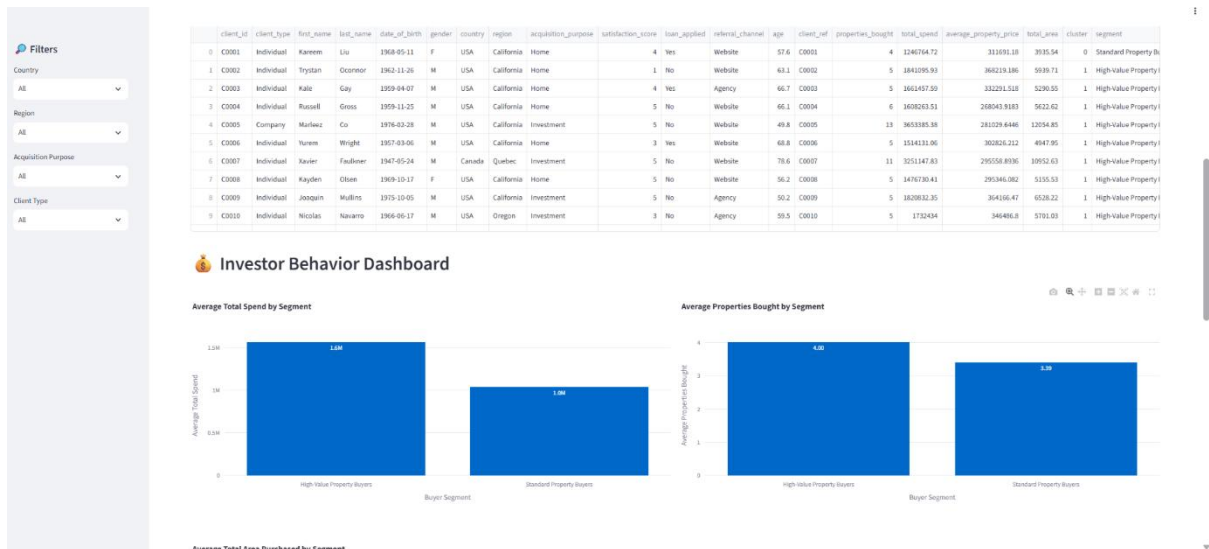


Figure 8: Stream lit Dashboard — Buyer Behaviour Analysis

3. Geographic Buyer Analysis:

This section presents the top regions according to the number of buyers, helping identify geographic concentrations within the dataset.

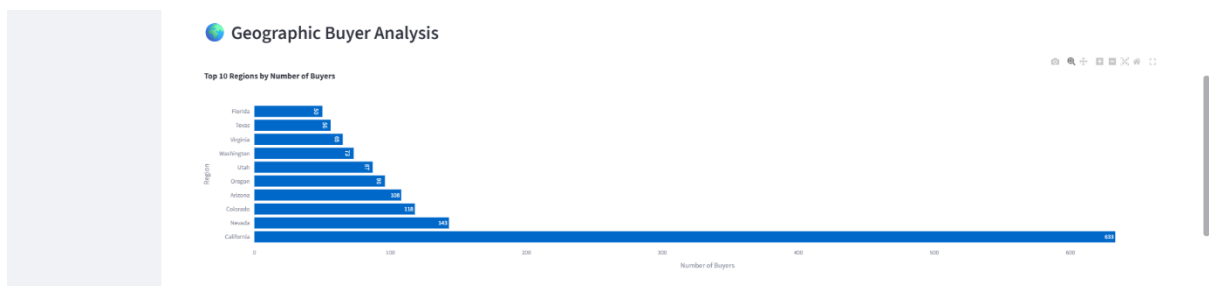


Figure 9: Stream lit Dashboard — Geographic Buyer Analysis

4. Segment Insights Panel:

This section provides a summary of the characteristics of each buyer segment, including buyer count, average age, satisfaction, property purchases, spending, average property price, and total area.

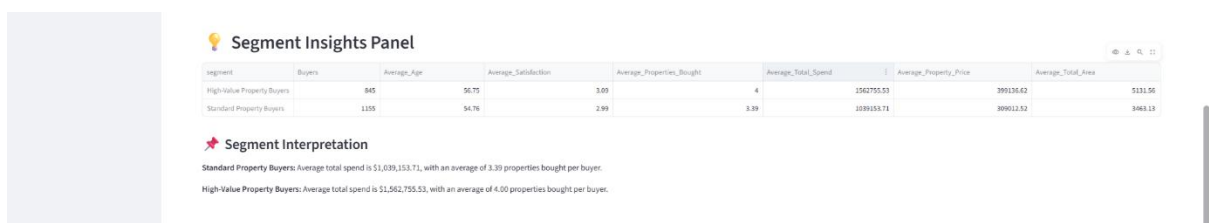


Figure 10: Stream lit Dashboard — Segment Insights Panel

The dashboard also provides filters for **country**, **region**, **acquisition purpose**, and **client type**. Users can select different filter values and dynamically explore the corresponding buyer segments and transaction patterns.

The Stream lit dashboard provides an accessible interface for exploring the machine learning results and can support data-driven decision-making in real estate marketing and customer analysis.

Recommendations

Based on the results of the buyer segmentation analysis, the following recommendations can be made for real estate organizations:

1. Develop segment-specific marketing strategies:

Real estate companies can create different marketing approaches for Standard Property Buyers and High-Value Property Buyers instead of treating all customers in the same way.

2. Provide personalized property recommendations:

High-Value Property Buyers can be provided with personalized recommendations based on their observed purchasing activity, preferred property characteristics, and transaction history.

3. Focus on geographic targeting:

Since the buyers are geographically concentrated in certain regions, regional marketing campaigns can be developed to target areas with a larger customer base.

4. Improve customer engagement:

Buyer characteristics and satisfaction scores can be used to design more relevant communication and customer engagement strategies.

5. Use transaction data for future analysis:

Regular analysis of property transaction data can help organizations monitor changes in buyer behaviour and identify emerging customer segments.

6. Integrate additional financial information:

Future versions of the system could include verified income, budget, financing amount, investment value, and other financial variables. This would allow more detailed investment profiling than is possible with the current dataset.

7. Continuously evaluate segmentation models:

Since the current Silhouette Scores indicate some overlap between the buyer groups, future implementations should test additional clustering algorithms and feature combinations to improve segment separation.

Limitations

Although the project provides useful insights into real estate buyer behaviour, it has several limitations.

1. The dataset does not contain direct income information, so buyer income or wealth cannot be measured.
2. The identified High-Value Property Buyers segment is based on observed transaction behaviour and should not be interpreted as a high-income group.
3. The Silhouette Score of the final K-Means model is relatively low, indicating that the identified clusters have some overlap.
4. The analysis is based on the available client and property transaction data and may not represent the entire real estate market.
5. Historical transaction data may not fully represent future buyer behavior or changing market conditions.
6. Additional financial and economic variables could improve investment profiling and segmentation accuracy.
7. The current segmentation identifies broad buyer groups and does not predict future purchases or investment returns.

Future Scope

The project can be further improved by incorporating additional data sources and advanced machine learning techniques. Future development can include verified income, property investment value, financing details, household information, property preferences, and economic indicators.

More advanced clustering algorithms such as DBSCAN, Gaussian Mixture Models, and other unsupervised learning techniques can be evaluated to determine whether more meaningful buyer groups can be identified.

The system can also be extended from descriptive segmentation to predictive analytics. Future models could predict the likelihood of future purchases, customer churn, preferred property categories, or potential investment behaviour.

The Stream lit dashboard can be enhanced with interactive geographic maps, time-based transaction trends, downloadable reports, and automated segment recommendations.

Finally, the model can be periodically retrained using new transaction data so that the buyer segments remain relevant as market conditions and customer behaviour change.

Conclusion

This project developed a machine learning-based approach for buyer segmentation and investment profiling in the real estate market. Client and property transaction data were combined to create a buyer-level dataset containing demographic, behavioural, and transaction-related features.

After data preprocessing and feature engineering, K-Means clustering was applied to identify groups of buyers with similar characteristics. The Elbow Method and Silhouette Score were used to evaluate the clustering structure. The final model identified two segments: **Standard Property Buyers** and **High-Value Property Buyers**.

The High-Value Property Buyers demonstrated higher average total spending, number of properties purchased, average property price, and total purchased area compared with Standard Property Buyers. The analysis also showed that both segments were primarily composed of individual buyers and that home or personal use was more common than investment as an acquisition purpose.

Hierarchical clustering was evaluated as a comparison model, but K-Means achieved a higher Silhouette Score and was therefore selected as the final model.

An interactive Stream lit dashboard was developed to present the segmentation results, buyer behaviour, geographic analysis, and segment insights. The dashboard also allows users to filter the results by country, region, acquisition purpose, and client type.

Overall, the project demonstrates how machine learning and interactive analytics can be used to identify meaningful patterns in real estate buyer behaviour and support data-driven market intelligence and customer targeting. However, the identified segments should be interpreted as broad transaction-based groups because the available data does not directly measure income or complete investment capacity.

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